

Snapdeal Analytics – Power BI

Short Internship Project Report

This project was completed as part of my Data Analytics internship. The objective was to analyze Snapdeal product data and use Power BI and DAX to identify useful patterns related to pricing, discounts, ratings, reviews, promotions and inventory risk.

1. Tools Used

Tool	Purpose
Microsoft Power BI	Data analysis and visualization
Power Query	Data cleaning and preparation
DAX	Measures, KPIs and calculated analysis
GitHub	Project files and documentation

2. Analysis Completed

Task 1 – Pricing vs Satisfaction

Compared product price and ratings and created measures such as estimated sales, category averages and a risk score.

Task 2 – Promotion Effectiveness

Compared promotional and non-promotional discounts. Baseline discount: 14.67%; promotion discount: 69.65%; promotional lift: 54.99 percentage points.

Task 3 – Discount vs Rating

Studied discount, rating and review volume. The calculated regression slope was approximately 0.01, indicating a very weak relationship between discount and rating.

Task 4 – Financially At-Risk Inventory

Created a KPI for financially at-risk inventory. The calculated value was 25.40%.

Task 5 – Trust-Weighted Rating

Compared simple average, review-weighted and trust-weighted ratings. The trust-weighted rating was 4.11.

Task 6 – Context-Aware Price Banding

Created Low, Medium and High price bands using the 30th and 70th percentiles, with the thresholds responding to the selected data context.

3. Key Findings

- Promotional products had a much higher average discount than non-promotional products.
- Higher-discount groups showed higher average review volume in the available data.
- Average ratings were around 3.1 across the review-volume bands used in the analysis.
- The discount-to-rating regression slope of about 0.01 suggests that discount level alone has very little linear relationship with rating.
- The financially at-risk inventory KPI was 25.40% based on the derived risk calculation.
- The trust-weighted rating was 4.11 using review volume and the estimated return component.

4. Limitations Due to Available Data

- A true 30-day promotional trend could not be calculated because the dataset was a product-level snapshot without sufficient historical observations.
- Actual sales figures were not available, so estimated sales were derived from available fields rather than transaction data.
- Actual inventory value and stock history were not available, so the financially-at-risk KPI uses an estimated stock-value measure.
- Actual return-rate data was not available, so the return component used in the trust-weighted analysis is a proxy.
- The discount-rating analysis shows association in the available data; it cannot prove causation without controlled or historical data.

5. Conclusion

The project provided practical experience in data preparation, DAX, Power BI visualization and business-oriented analysis. Where the available dataset did not support an exact analysis, proxy measures were used and their limitations were documented.

Project files: Power BI file, DAX calculations, task screenshots and dataset link.